

Cunxin Huang

PhD Student | Industrial AI / Benchmarking / Evolving Systems / AI for Science

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Profile

- Target roles: **AI training and applications**, focusing on industrial AI, AI for Science, and algorithm evolution/benchmarking.
- Applied mathematics PhD student at PolyU researching **DFO**; familiar with LLM fundamentals and agent frameworks.
- **Agent tool use, code evolution, and benchmark/evaluator systems** for measurable, reproducible algorithm improvement.

Education

- 2022 – Present • **The Hong Kong Polytechnic University, Applied Mathematics, PhD Student**
Research: derivative-free optimization; supervisors: Prof. Zaikun Zhang, Prof. Xiaojun Chen; **HKPFS** awardee.
- 2018 – 2022 • **Jilin University, School of Mathematics, BSc**
Information and Computing Science (Computational Mathematics); **rank: 2/109**.
Selected for the “Qiusi Siyuan” Student Development Program (7th cohort).

Internships

- Aug 2026 – Present • **Tencent TEG, Model Department, Post-Training Team Intern**
- May 2026 – Present • **Yuanmu Intelligence, Algorithm Intern (Industrial AI / Production Scheduling)**
Python, LLM/Agent, code evolution, benchmark/evaluator
 - Built an industrial algorithm evolution system with candidate generation, benchmarking, held-out evaluation, and version tracking.
 - Refactored core scheduling operators and evaluation interfaces to support agents iteratively generating algorithms for new requirements.

Skills and Experience

- AI • Agent/RAG, tool use, heuristic learning, code evolution, benchmark/evaluator systems
- Engineering • Python, MATLAB, Git/Linux; derivative-free and numerical optimization; Chinese and English

Projects

- 2026 – Present • **Terminal-Bench Science: Scientific Computing Agent Benchmark**
[Website / Leaderboard] [GitHub] [vo.1.o] [Announcement]
 - Co-developed **4 scientific computing tasks** spanning black-box optimization, guided-wave localization, and digital twins; all merged upstream and included in vo.1.o.
 - **OpenAI GPT-6 Astra**: benchmark used and prominently featured at launch. [Launch page]
 - **Anthropic Claude Fable 5.1**: benchmark used and prominently featured at launch. [Launch page]
- **OptiProfiler Evolve: Automated DFO Solver Evolution**
Python, DFO, code evolution, benchmarking
 - Built a solver evolution loop on OptiProfiler’s benchmark/evaluator, supporting candidate search, staged evaluation, and held-out validation.
- 2024 – Present • **OptiProfiler: DFO Benchmarking and AI Agents** [Platform] [GitHub] [Agent]
Python, MATLAB, numerical optimization, RAG, benchmarking
 - Primary developer; built a DFO solver benchmarking platform and agent tool workflows for cloud execution, result analysis, and diagnostics.

Selected Research (more on my website)

- 2026 • **OptiProfiler: a platform for benchmarking optimization solvers** (with T. M. Ragonneau and Z. Zhang)
 - **Non-convergence Analysis of Probabilistic Direct Search** [arXiv:2606.01320] (with Z. Zhang)
 - **A Unified Series Condition for the Convergence of Derivative-Free Trust Region and Direct Search** (with Z. Zhang)
- 2023 • **Error Analysis of Finite Difference Scheme for American Option Pricing** (*J. Comput. Appl. Math.*)

Honors and Recognition

- 2026 • **SIAM OP26 Student Travel Award**; invited talk at SIAM OP26
- 2025–2024 • Invited talks at ICCOPT / ISMP; ORSC Operational Research Application Award (team member); PolyU Best Presentation Award
- 2023–2021 • Outstanding Presentation Award, ORSC Mathematical Programming Graduate Forum; MCM/ICM Finalist (top 2% worldwide); Qi An-Xin Scholarship
- 2022–2019 • **Hong Kong PhD Fellowship Scheme (HKPFS)** (approx. 200/year); National Scholarship; Zhao-Yue Scholarship

References (available upon request)

- **Michael A. Saunders**, Professor (Research) Emeritus, Stanford MS&E / ICME; SIAM Fellow [Website]
- **Philippe L. Toint**, Emeritus Professor, University of Namur; SIAM Fellow; former MOS Chair [Website]